

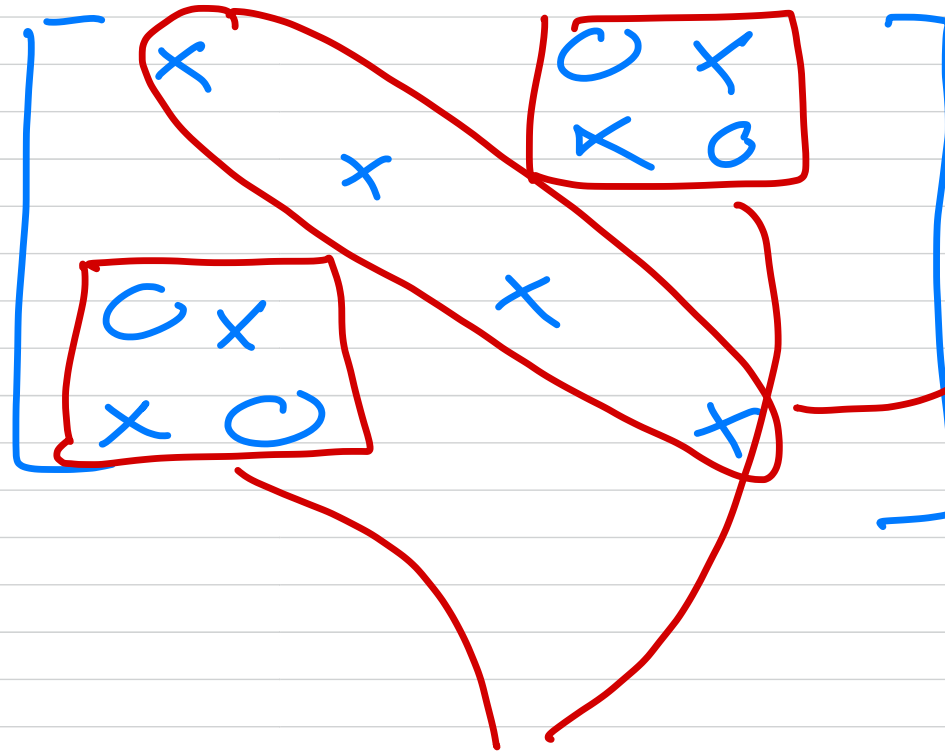
FYS5419/9419 March 4



FYS5419/9419 March 4

Two-qubit $\mathcal{H}$

$$\hat{\mathcal{H}} =$$



$$\epsilon_{11}(\mathbb{I}_1 \otimes \mathbb{I}_2)$$

$$+ \epsilon_{z1}(\hat{z}_1 \otimes \mathbb{I}_2)$$

$$+ \epsilon_{1z}(\mathbb{I}_1 \otimes \hat{z}_2)$$

$$+ \epsilon_{zz}(\hat{z}_1 \otimes \hat{z}_2)$$

$$V_x(x_1 \otimes x_2) + V_z(\hat{z}_1 \otimes \hat{z}_2)$$

$$|00\rangle$$

$$|01\rangle$$

$$|10\rangle$$

$$|11\rangle$$

$$X_1 \otimes X_2 = X_1 X_2$$

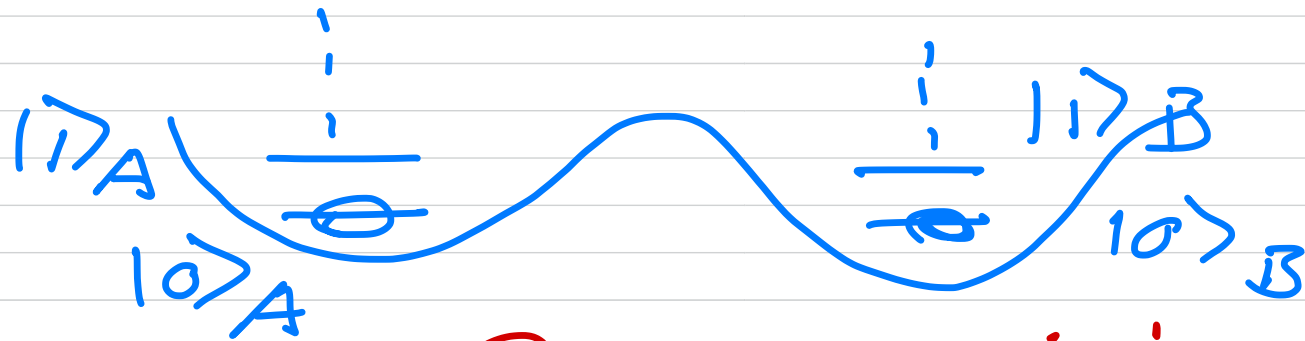
$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$X_1 X_2 |00\rangle = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

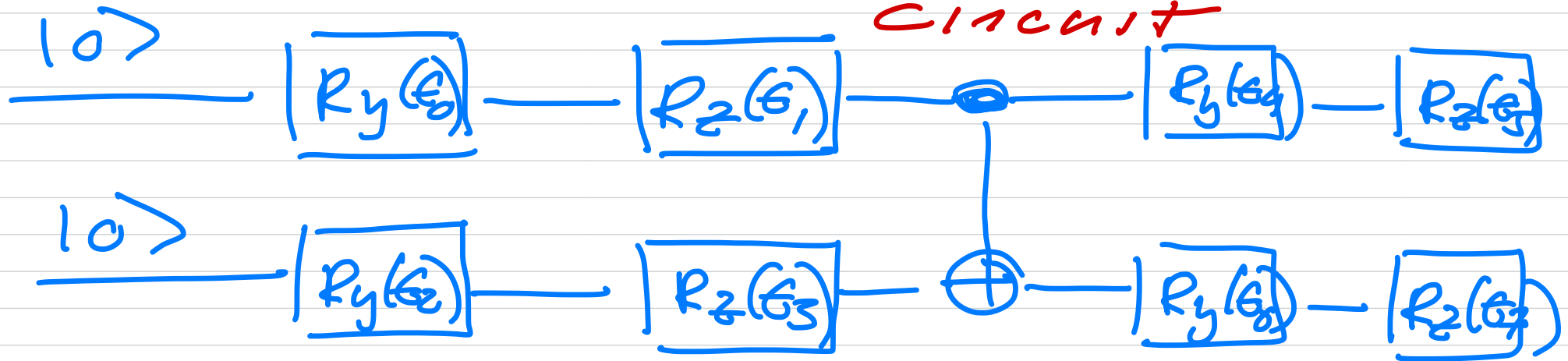
$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle$$

$$X_1 \rightarrow H Z_1 H = X_1$$

Ansatz for VQE



Parameterized Quantum Circuit

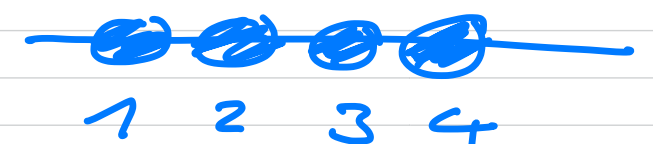


$$\Theta = \{ \theta_0, \theta_1, \theta_2, \dots, \theta_7 \}$$

$$N=2 \quad \vee \quad N=4$$

$$\# \text{ configs} = \binom{8}{4}$$

P  $\sum S_z = 1/2 \quad (\uparrow)$

P  $-\sum S_z = -1/2 \quad (\downarrow)$

$$N=4$$

↓
Total spin projection

$$S_z = \sum_{i=1}^N S_{z_i} = -2$$



$$J_z = \sum_{i=1}^N J_{z_i}$$

$$[H, J_z] = 0 \quad \wedge \quad [H, J^2]$$

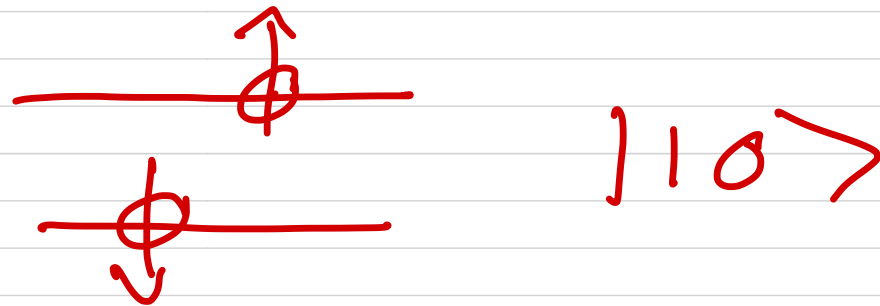
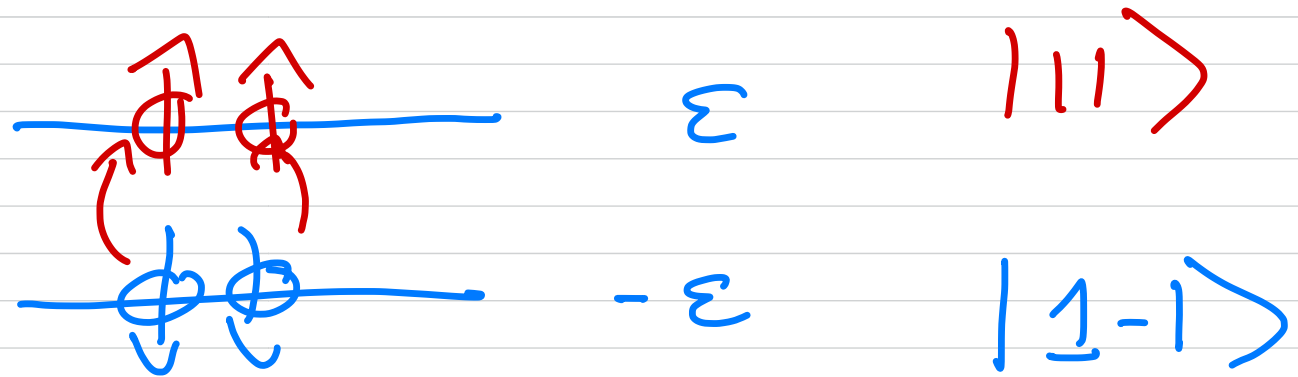
$$J^2 = J_x^2 + J_y^2 + J_z^2$$

$$J = 2 \quad J_z = -2, -1, 0, 1, 2$$

$$|J=2, -2\rangle, |2, -1\rangle, |2, 0\rangle, |2, +1\rangle, |2, 2\rangle$$

$$|4\rangle = \alpha |2, -2\rangle + \beta |2, -1\rangle + \gamma |2, 0\rangle + \delta |2, 1\rangle + \mu |2, 2\rangle$$

$$N=2$$



$$|4\rangle = \alpha |11\rangle + \beta |10\rangle + \gamma |1-1\rangle$$

$$N=2, \quad 2^2=4 \quad \{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$$

$$J=2$$

$$2^3 = 8 \quad \left\{ \begin{array}{l} |000\rangle, |001\rangle, |010\rangle, |100\rangle \\ \quad \quad \quad \uparrow \\ \quad \quad \quad [1] \otimes [1] \otimes [1] \quad \dots \quad |111\rangle \end{array} \right\}$$

$$J=1 \quad (N=2)$$

$$|1-1\rangle, |10\rangle$$

$$|11\rangle$$

$$\langle 1-1 | \text{ or } |1-1\rangle$$

$$\langle 1-1 | \text{ or } |10\rangle$$

$$\langle 1-1 | \text{ or } |11\rangle$$

$$\langle 10 | \text{ or } |10\rangle$$

$$\langle 10 | \text{ or } |11\rangle$$

$$\langle 11 | \text{ or } |11\rangle$$

$$\begin{array}{c}
 \langle 1-1 | \\
 \langle 10 | \\
 \langle 11 |
 \end{array}
 \begin{array}{ccc}
 |1-1\rangle & |10\rangle & |11\rangle \\
 \left[\begin{array}{ccc}
 -\epsilon & 0 & V \\
 0 & W & 0 \\
 V & 0 & \epsilon
 \end{array} \right]
 \end{array}$$

$$\begin{array}{c}
 \langle 10 | \\
 \langle 1-1 | \\
 \langle 11 |
 \end{array}
 \begin{array}{c}
 |10\rangle \\
 \left[\begin{array}{ccc}
 W & 0 & 0 \\
 0 & -\epsilon & V \\
 0 & V & \epsilon
 \end{array} \right]
 \end{array}$$

$\langle 10 | H | 10 \rangle = W$

VQE with
1 qubit

$$J = 2$$

$$N = 4$$

$$\begin{array}{ccccc}
 |2-2\rangle & |2-1\rangle & |20\rangle & |2+1\rangle & |22\rangle \\
 \begin{bmatrix}
 -2\varepsilon & 0 & \sqrt{6}V & 0 & 0 \\
 0 & -\varepsilon + W & 0 & 3V & 0 \\
 \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\
 0 & 3V & 0 & \varepsilon + 3W & 0 \\
 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon
 \end{bmatrix}
 \end{array}$$

with 3 qubits $2^3 = 8$

$|2-2\rangle$ $|20\rangle$ $|22\rangle$ $|2-1\rangle$ $|21\rangle$

-2ε	$\sqrt{6}V$	0
$\sqrt{6}V$	$4W$	$\sqrt{6}V$
0	$\sqrt{6}V$	2ε

2 quanta

C	C	C
C	C	C

$-\varepsilon+W$	$3V$
$3V$	$\varepsilon+3W$

1 quantum case